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Subsurface geometry and structural evolution of La Gomera island based on gravity data

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ABSTRACT

We hereby present a new Bouguer gravity map of the La Gomera island (Canarian Archipelago), which is analysed and interpreted by means of a 3-dimensional inversion, in order to contribute to the knowledge of the structural setting of this volcanic island and its evolutionary history. A land gravity data set covering the whole island of La Gomera is used in combination with offshore measurements to achieve a better determination of the gravity field in areas near the coasts.

The study of this map let us to shed some light on the hypothesis established about the volcanism of this island. Moreover, it shows the information that is hidden from a geological surface exploration, modelling deep sections of the crust in La Gomera, which have been unknown until now.

A first interpretation of the Bouguer gravity anomaly is achieved from 1) the residual gravity map calculated by removing a regional component and 2) the total horizontal gradient of the gravity. These residual and derivative maps allow us to identify the horizontal location and borders of the shallowest gravity sources. This provides a useful tool to study the structures associated to the latest periods of the volcanism in the area. Moreover, the information so obtained supports the hypothesis about the migration of volcanic activity towards the south of the island.

Subsequently, an inversion process is carried out looking for the 3D-modelisation of the sources of the observed gravity field, which provides a comprehensive view of the structures in volcanic environments. The inversion technique used is based on a genetic algorithm (GA) applied upon a prismatic partition of the subsoil volume, and adopting a priori values of density contrast (positive and negative). The main advantage of this method is that let us to model deep and shallow bodies which exhibit very different geometries and density contrasts. So, results indicate that this inversion strategy can be very effective for characterization of volcanic structures, improving the information from previous geologic and volcanologic studies. The inversion model obtained shows correlation between several sources of the gravity field and the volcanic units associated with the growth of La Gomera Island. The main gravity source of this model is associated with the oldest unit, called the Basal Complex. This unit corresponds to the first submarine growth stage and it is modelled as the most important and deepest high density structure. According to previous geological studies, the following edifice (Old Edifice) was also submarine in its initial phases, later being represented by a wide basaltic shield volcano. The original location and morphology of this Old Edifice is deduced from the distribution of positive density contrasts that appears in the model. Moreover, other gravity field sources are identified and associated to several feeding systems of this stage of the volcanism in La Gomera. The shallowest sections of the model let us recognise the distribution of light material inside the Vallehermoso caldera, surrounded by high density structures. This gives us some insight into the internal structure and morphology of the caldera, pointing to a vertical collapse origin followed by erosion and other destructive processes. Finally, other conclusions are obtained from the correlation found between the sources of the gravity field and the migration of the volcanic activity towards the southern area of the island.

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1. Introduction

The island of La Gomera occupies a central position in the Canarian archipelago (Fig. 1), being located between Tenerife and the western islands of La Palma and El Hierro. La Gomera shows geological characteristics that differ from those islands in the vicinity, due to its much older age and its lack of Quaternary activity (Abdel-Monem et al., 1971). Thus, La Gomera is the only one that can be considered a non-active island in the Canarian archipelago in spite of it being surrounded by three active islands (Tenerife, El Hierro and La Palma) (Carracedo, 1999).

Several previous geochronological, volcanic and morphological studies have been carried out in La Gomera (Cendrero, 1970, 1971;

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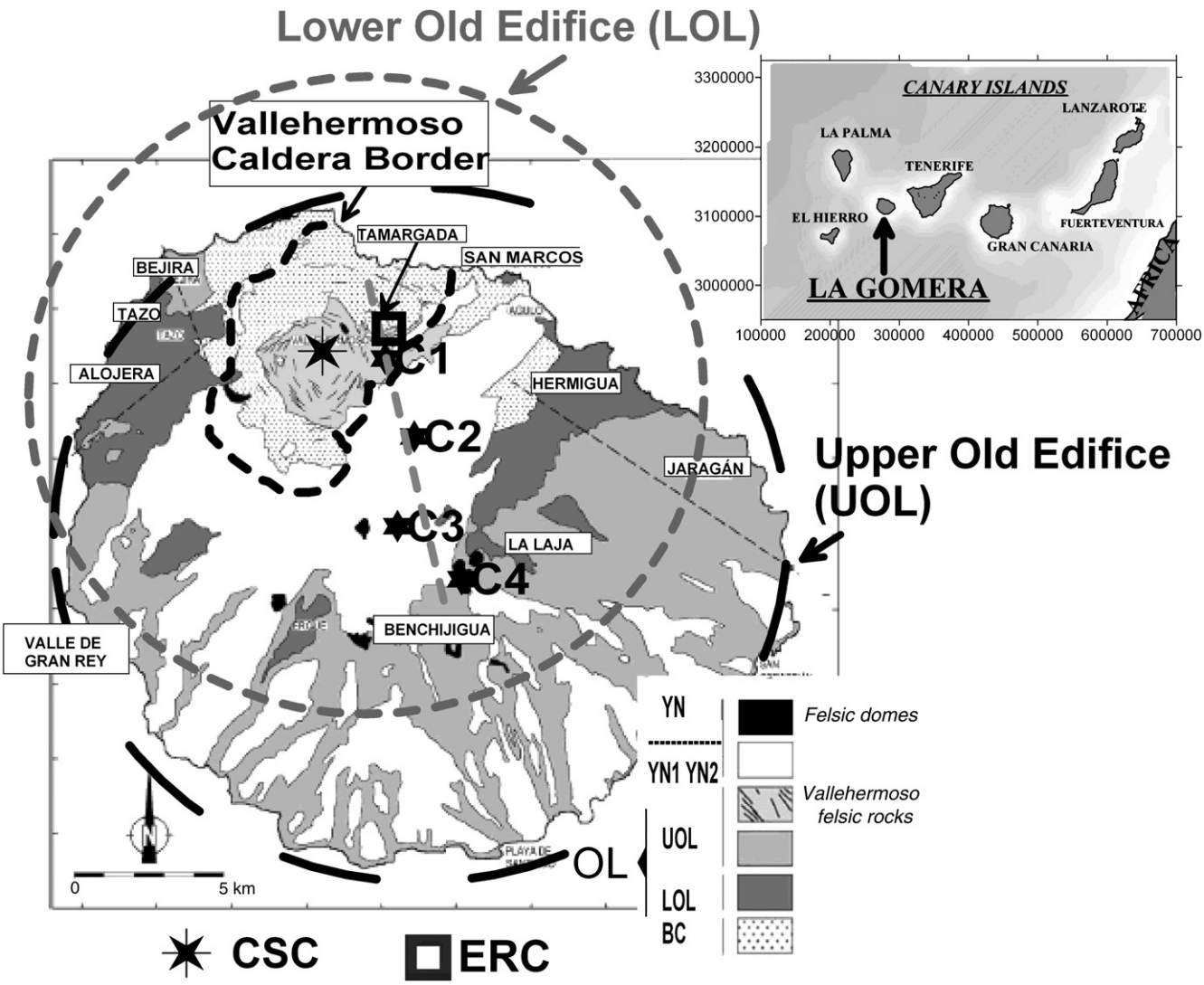

Fig. 1. Simplified geological map of main stratigraphical units of La Gomera (adapted from Ancochea et al., 2006). Locations of the LOL (gray dotted circle) and UOL (black dotted circle) according to Ancochea et al. (2006), the estimated Cone Sheet Centre (CSC) and Eastern radial centre (ERC) (Ancochea et al., 2003). Distribution of the four volcanic centres (C1 to C4) defined by the convergence of dikes ascribed to the respective swarms (stars). Dotted gray line indicates the migration of volcanic activity from Ancochea et al. (2008). The inset figure shows the geographical location of La Gomera Island at the Canarian Archipelago.

Rodríguez-Losada, 1987, 1988; Ancochea et al., 2003, 2006, 2008; Rodríguez-Losada and Martínez Frías, 2004; Paris et al., 2005; Llanes et al., 2009; etc.). To date, there are no local scale studies about the inversion of potential fields, which let us obtain information about the crustal structures of this volcanic island from the modelling of the sources of the observed field.

The use of gravity modelling plays an important role in the study of volcanic ocean islands, where complex structures are usually present (e.g., Rymer and Brown, 1986; Minshull and Brozena, 1997; Malengreau et al., 1999; Kauahikaua et al., 2000; Klingelhofer et al., 2001; Cella et al., 2007; Gottsmann et al., 2008; Paoletti et al., 2009). With regard to the Canarian archipelago, detailed land gravity studies have been achieved in all islands except La Gomera, obtaining the corresponding crustal models (e.g, Araña et al., 2000; Camacho et al., 2001, 2009; Montesinos et al., 2005, 2006; Gottsmann et al., 2008). Thus, with this work and the preceding ones, the gravity inversion study for the entire archipelago could be complete, with a crustal model for each of the islands.

La Gomera is a single large, complex and long-lived volcanic edifice with three main growth stages distinguishable: the Submarine Edifice (SB), the Old Edifice (OL) and the Young Edifice (YN). However, there are different hypotheses about the volcanological evolution and stratigraphic models of the island from several authors (see Ancochea et al., 2006). Among these controversies, the Vallehermoso area and its surrounding environment represent one of the more interesting features of La Gomera, which has lead to several theories related to its origin and evolution (e.g. Rodríguez-Losada and Martínez Frías, 2004). Ancochea et al. (2003) estimate the existence of an ancient single dome-shaped shallow magma chamber situated 1650 m b.s.l. in this area. Whilst Rodríguez-Losada and Martínez Frías (2004) suggest a collapse of the magma chamber, Ancochea et al. (2006) establish that the caldera-shaped morphology is of erosional origin, and Cubas et al. (2002) suggest a giant landslide which formed the Vallehermoso embayment.

The usefulness of the gravity inversion to provide insight into collapsed calderas (eg. Toba, Teide caldera) has been shown in many studies (e.g. Masturyono et al., 2001; Gottsmann et al., 2008). In particular, in the case of the Vallehermoso caldera it is possible to observe some analogies with the results obtained in the neighbouring island of Tenerife, where the gravity studies support the hypothesis of a vertical collapse as the origin of the Teide caldera (Gottsmann et al., 2008).

Taking into account previous studies, we conducted a 3D gravity inversion of La Gomera, to provide some insight into its subsurface structures. In this paper, in Section 2, we look over the geological and

volcanological studies of La Gomera island, pointing out several structural aspects that are important for the following gravity study. In Section 3, we explain the gravity and topographic data used and how they are analysed to estimate the corresponding noise level. In Section 4, a first interpretation of the gravity data is made from the local components and the total horizontal gradient of the gravity field. With this gradient map it is possible to identify shallow structures that can be hidden because of the strength of the gravity signal of the deeper sources. Finally, we apply our inversion method to the gravity map, this way obtaining a model of the crustal structures of the island of La Gomera. This model allows us to identify several structures which are explained successfully from the volcanic history of the island. The inversion technique and the corresponding results are described in Sections 5 and 6.

2. Geological and volcanological setting of La Gomera

La Gomera island has a roughly circular surface of about 372 km^2 with a 24 km east–west diameter and a north–south length of 20 km. The centre of the island is occupied by an undulating upland with a maximum elevation of 1487 m at Alto Garajonay (Fig. 2). The most distinctive morphological feature is the intense degree of erosion in all formations, with deep, vertical-walled valleys cutting the island radially and where the tabular successions of basalts can be seen. The emerged part represents only 4.5% of the whole edifice, about 11,000 km^3 (Paris et al., 2005).

Among the authors who have studied the geological and volcanological sceneries of the island, Ancochea et al. (2006) establish a new model for the general volcano-stratigraphy and geochronology of the island. This evolutionary model distinguishes three main growth stages for the island (Fig. 1): the Submarine Edifice (SB), the Old Edifice (OL) and the Young Edifice (YN). In turn, these last two stages, corresponding to subaerial growth, are subdivided into: Lower Old Edifice (LOL), Upper Old Edifice (UOL), Young Edifice-1 (YN-1) and Young Edifice-2 (YN-2).

According to Ancochea et al. (2006), the evolution of the volcanic island was as follows. The oldest edifice corresponds to the submarine growth stage of the island and its hypabyssal roots. This edifice, the so-called Basal Complex (BC) (Bravo, 1964), crops out exclusively in the north of the island. It appears in Hermigua and Vallehermoso areas, mainly at the south, west and north into the Vallehermoso Caldera (Fig. 1). This complex is analogous to those existent in La Palma (Staudigel and Schmincke, 1984) and Fuerteventura (Le Bas et al., 1986). The BC rocks (mafic plutonic, submarine volcanic, very scarce marine sediments traversed by a dense network of mainly basic dikes) appear markedly deformed, particularly in the NW sector where well-developed ductile deformation structures are seen (Cendrero, 1971).

The geochronology of this early stage is a question still unsolved (Ancochea et al., 2006). Some authors establish that this submarine stage was built up between 20 and 15 Ma (Abdel-Monem et al., 1971) and it was followed by an important break in the activity (about 4 Ma), deep erosion and destruction of the edifice.

About 10.5 Ma ago, the main present-day edifice, the OL (10.5–6.4 Ma) emerged. It was also submarine in its initial phases, and it took place upon the remaining and relatively deep hypabyssal roots of the preceding edifice. It is not possible to determine the original morphology or the precise location of the OL from geological information. Therefore, a structural subsurface model from gravity inversion can shed light on this matter. Nevertheless, Ancochea et al. (2006) inferred that the submarine formation was centred at the north of the younger subaerial ones, taking into account the existence of the exposures of the submarine rocks restricted to the northern area and the later southwards migration of the activity.

At present, the OL reaches a maximum altitude of 1100 m near the centre and extends northwest, west, south and east of the island. It rests with unconformity over the BC, sometimes separated by sedimentary deposits generated when the BC was eroded. The upper boundary of the OL with the younger units is quite unclear. Two different main stages are distinguishable in this edifice (Ancochea et al., 2006). The

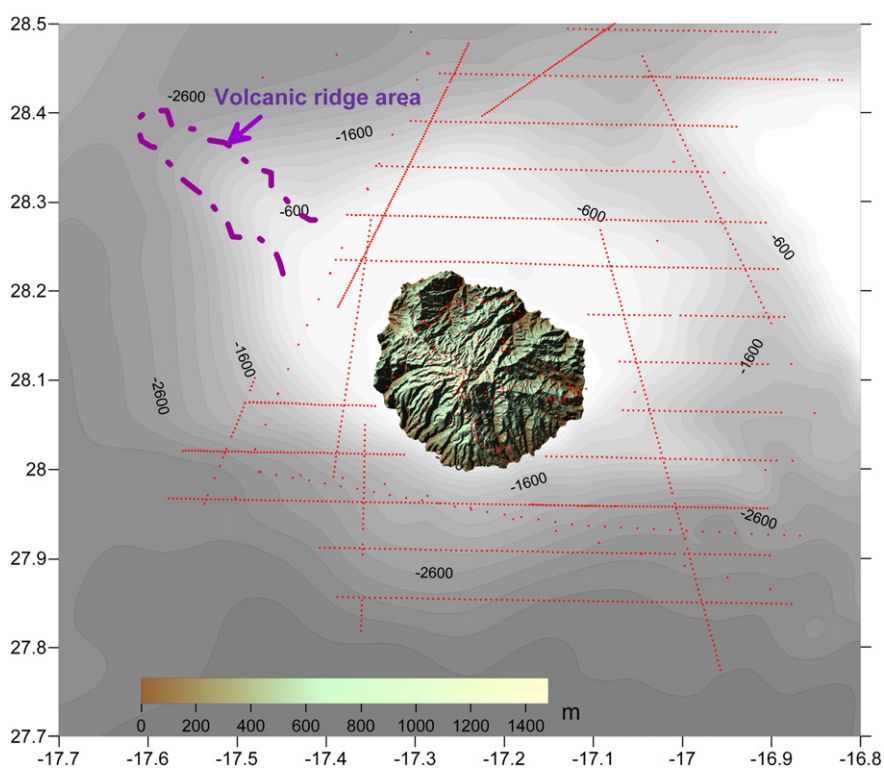

Fig. 2. Ortographic view of the digital terrain model of La Gomera and distribution of gravity stations. The centre of the island is occupied by an undulating upland with a maximum elevation of 1375 m (Alto de Garajonay). The bathymetry is shown as contour isolines. The purple dotted line at La Gomera northwest flank indicates the submarine ridge without an onshore extension (Llanes et al., 2009). Its crest is covered by cone-like shape structures and it is interpreted as of volcanic origin. The geographic coordinates are given in degrees.

first stage was represented by a large, approx. 22,000 m wide basaltic shield volcano (gray dotted circle in Fig. 1), fairly well represented in the NW and NE sectors, at the vicinity of Alojera and Hermigua, respectively. It is also exposed at minor vents in the main, more excavated southern ravines and in San Marcos (Fig. 1). The centre of the original construction would be situated in the vicinity of Vallehermoso, about 8000 m north of the central zone of the present island, being originally larger in size than its current geometry. The northern rim of the emergent edifice would be extended approximately 5000 m further offshore the present northern coastline. According to these authors, a volcanic edifice of such dimensions may have reached a height of 1300 to 1900 m. Several lateral collapse events, such as the Tazo and San Marcos avalanches (Fig. 1), occurred during this time and were responsible for the removal of an important part of its northern flank. Whatever mass vesting events took place and led to the creation of the present morphology of La Gomera, are no longer recognisable offshore (Acosta et al., 2003).

During the second growth stage, corresponding to the UOL, the activity migrated southwards. The UOL was originally larger in size than the present island. A 25,000 m wide composite volcano arose covering part of the remaining earlier shield volcano, and centred near the present centre of the island (black dotted circle in Fig. 1) (Ancochea et al., 2006). On the one hand, the UOL consists of basalts that were erupted between 8.6 and 6.4 Ma ago, the best exposition of this volcanic unit appearing in the eastern sector of the island (Jaragán area). On the other hand, the main growth of this edifice is associated with felsic activity on the northern part of La Gomera, which is studied by Ancochea et al. (2003, 2006) and by Rodríguez-Losada and Martínez Frías (2004), among others. This activity is located around the Vallehermoso and Tamargada areas and it implies a felsic cone sheet dike complex, caldera formation and dome intrusion.

Ancochea et al. (2003) identified three different dike swarms, intruded possibly during two separate time episodes. The first felsic episode comprises a group of felsic domes, thick flows and pyroclastics and the eastern radial swarm which is centred in the vicinity of Tamargada (ERC in Fig. 1). During the second felsic episode (the "Vallehermoso Felsic Rocks"), about 7.5 Ma ago, the activity migrated some 2500 m southwest, giving rise to a second radial swarm and the Vallehermoso Cone Sheet Swarm (with centre in CSC indicated in Fig. 1). From the geometry of the cone sheet, these authors estimated the existence of an ancient single dome-shaped shallow magma chamber situated some 1650 m below the present sea level. In addition to a mid-to-upper Miocene felsic cone sheet dike complex, Rodríguez-Losada and Martínez Frías (2004) located and studied a caldera collapse and dome intrusion in the Vallehermoso area. Based on detailed field and structural evidence, these authors suggested that the cone sheet shows an almost circular geometry at the surface, truncated to the NW sector. After the cone sheet dike complex formed, decompression of the interior of an assumed shallow magma chamber occurred, due to the subsequent migration of felsic magma throughout the dikes of the cone sheet. This last event caused the roof of the magma chamber to collapse, leading to the formation of a 3000 to 4000 m wide caldera. A breccia deposit formed during the collapse and filled a depressed central area that was finally affected by an episode of magmatic resurgence, leading to a later, almost circular dome intrusion (Rodríguez-Losada and Martínez Frías, 2004). The association of cone sheets, breccia deposits and domes constitutes the so-called "Trachytic-Phonolitic Complex" (T-PhC). Previously, Rodríguez-Losada (1988) and Cueto et al. (1994) had already suggested the possible existence of a caldera in association with the cone sheet complex. However, Ancochea et al. (2006) disagree with these authors and the assumption about the formation of a collapse-caldera related to the T-PhC. The hypothesis of Ancochea et al. (2006) establishes that the caldera-shaped morphology, visible at present in Vallehermoso, is of erosional origin and much younger (<4 Ma), because the horizontal thick flows of the YN are exposed at the wall so generated. Other authors (Paris et al., 2005) define this area as a 10,000 m wide embayment, open to the north and filled by post-shield volcanism, which is supposed to be a giant landslide embayment (the 'Garajonay landslide').

Finally, the last volcanic unit, the YN, corresponds to a more recent large edifice built up from 5.7 to 4 Ma. The lava flows of this younger edifice never covered the entire island, only the centre and the south area and filled deep ravines in the north. Several authors consider different dating and stages for this Edifice (Bravo, 1964; Cendrero, 1971; Cantagrel et al., 1984; etc.). In a more recent study, Ancochea et al. (2006) identify two stages: in the early phase, YN-1, lava flows were erupted from the central area of the island and descended essentially south and southwestwards; and in the second stage, YN-2, abundant lava flows spread all over, covering a vast extension and filling deep valleys reaching out to the W, N and NE peripheries. The last 2 Ma is a period of total quiescence, with no Quaternary activity at all.

During the building of these insular volcanic edifices that formed the present La Gomera, intrusion of magma caused the development of an enormous amount of dikes that, step by step, constituted the main framework of the hypabyssal roots of these edifices. The knowledge and analysis of their distribution allowed to infer several aspects about the volcanic history of the island (Ancochea et al., 2003; Hampton and Cole, 2009). Ancochea et al. (2008) identify four volcanic centres defined by the convergence of dikes ascribed to the respective swarms, indicating a migration of volcanic activity from the first centre to the north towards the fourth, which is the youngest and southernmost one (C1 to C4 in Fig. 1). On the basis of radiometric ages of the dikes and of the rocks they cross, these authors closely relate the first and second swarms to the Old Edifice and associate the last two swarms with the Young Edifice. Therefore, it is shown that La Gomera is not the result of the building of a single large volcanic edifice, but instead has gradually grown southwards with a very slow displacement of the magmatic focus (from 9 Ma to 4 Ma).

3. Gravity data

We used a total of 82 new land gravity stations covering the island, with a distribution conditioned by the rugged topography (Fig. 2). The gravity survey was made with a LaCoste&Romberg mod-G gravimeter and the respective coordinates of the data points were obtained by differential GPS. The gravity data set was corrected with the tidal model of the Canaries calculated from the tidal gravity observations made in several islands of the archipelago (e. g., Arnoso et al., 2006, in press). To achieve a denser distribution of land gravity stations, we added the gravity values of 192 stations provided by the National Geographic Institute of Spain (Fig. 2). To guarantee the merging of these two data sets, some of these gravity stations were repeated in our survey. The adjustment of gravity data corrected by drift, jumps and tides, produced residuals with a standard deviation of 0.084 mGal ($1\ \mathrm{mGal} = 10^{-5}\ \mathrm{ms}^{-2}$).

We also used marine gravity data provided by the U.S. Geological Survey (USGS) and collected from a cruise aboard the Starella Research Vessel in 1987 (Folger et al., 1990). These data correspond to up to 10,000–15,000 m offshore and have a minimum separation of 1000 m between contiguous observations (Fig. 2). Using the marine data, we can better identify the offshore prolongation of land gravity anomalies and we can detect (following section) if there is a regional trend in the gravity map. All the gravity values are referred to IGSN 1971.

To calculate the terrain correction for the gravity data, a digital terrain model (Fig. 2) of 1:25,000 scale provided by the National Geographic Institute of Spain is used. This land data set is completed with ETOPO1 (National Geophysical Data Center, USA Amante and Eakins, 2008) bathymetric data.

The terrain correction is applied up to 40,000 m away from each station, according to the regularity of the effect beyond this distance.

To select a suitable average terrain density, we choose a value that produces a smaller distortion of the gravity anomaly field. We test several density values to calculate the corresponding gravity anomaly map and a covariance analysis was repeated for all of them. For each density value, the root mean square error (rmse) of the gravity map is calculated applying the following iterative approach. Using the autocorrelation analysis of the data and calculating a suitable analytical covariance function (e.g. Barzaghi and Sanso, 1983), the usual formulae for least-squares prediction (Moritz, 1980) give us a first estimation of the signal at this level of covariance and the corresponding prediction error matrix. By considering the residual values (observed data minus estimated signal), we repeat this process again to detect, if possible, correlated signals of minor amplitude (levels). This procedure ends when no further signal level can be detected.

Finally, by eliminating the sum of the signal levels detected from observed data, the distribution of the resulting residual values displays the non-correlatable noise (e.g. Montesinos et al., 2005). With the rmse corresponding to these residual data, we check the quality of the data for the density value used for the terrain correction. So, looking for the minimum rmse, we considered a mean value of 2560 kg/m^3 for the density of the terrain masses and a value of 1027 kg/m^3 for the seawater. This terrain density is as close to the value of 2570 kg/m^3 as can be obtained by Nettleton's method. Moreover, this value is near the density values calculated for the other islands of the archipelago, which have a similar volcanism: 2500 kg/m^3 for Fuerteventura (Montesinos et al., 2005), 2510 kg/m^3 for El Hierro (Montesinos et al., 2006), 2320 kg/m^3 for Tenerife (Araña et al., 2000), etc. This value will represent an average density for the various regions of the island and it will be reflected in the shortest wavelength gravity anomalies. In this same way, the results of the inversion process can be influenced by an excess or lack of terrain correction in some areas.

There aren't any measurements of the densities of the rocks of La Gomera. However, from the geological sketch of La Gomera, it is possible to consider the value of mean terrain density used in this work in agreement with the geology of the island. This sketch shows the existence of the Basal complex with deep sea sediments and lavas, alkaline plutonic rocks, syenitic intrusions (alkaline gabbros and syenites s.s., old basalts, volcanic agglomerates, traquitic and fonolitic complex, subrecent basalts and domes and salic lava flows).

The Bouguer gravity anomaly calculated is shown in Fig. 3. Using the aforementioned process, three levels of gravity signal are defined by their covariance functions (Fig. 4). The first level is calculated from land and marine gravity data and the second and third levels (shorter wavelength) only from land data. With this process, the successful homogeneity between the data sets is shown because if there were a discrepancy between land and marine data, it would not be possible to achieve the covariance analysis and the estimation of the signal. The final residual of the estimated signal with respect to the observed data is the non-correlated noise, which shows a standard deviation of 1.77 mGal. The value of this residual shows the quality of the data, which is quite enough for this study. This noise may be related to observation itself and to very local anomalies or to uncertainties in the previous corrections (such as terrain corrections using a unique density value). The error matrix of the estimated signal calculated from the covariance analysis will be considered as an error matrix for the data, E_{ss}, which is used in the inversion process that follows.

The gravity anomaly map obtained (Fig. 3) ranges 130 mGal and shows maximum values at the northern section of the island and minimum values in the eastern offshore. The main maximum of gravity, which appears where the Basal Complex outcrops (Fig. 1), has its offshore prolongation towards the north. The marine gravity data helps to mark out the centre of this high anomaly near the northern coast.

The eastern zone offshore of the gravity map is dominated by a strong minimum of gravity situated over the saddle in the trough between La Gomera and Tenerife, the La Gomera promontory. Previous seismic works (Canales et al., 2000) associate travel-time delays with the La Gomera promontory and they suggest that this feature may represent large accumulations of lava flows and volcaniclastics deposited during the construction of the shield massifs of Teno and Roque del Conde, in Tenerife Island, and La Gomera during the Upper Miocene. This could be the source of the low gravity values observed in this area.

On the northwest corner offshore of the La Gomera insular slope, an elongated anomaly of maximum gravity is predicted (Fig. 3), despite there being a lack of gravity data. This lightly appreciated feature of the gravity map comes from the existence in this area of a ridge with its crest covered by 20 cone-like shape structures interpreted as of volcanic origin (Fig. 2) (Llanes et al., 2009). This gravity anomaly coincides with the existence of a linear magnetic anomaly (Catalan et al., 2003) that Llanes et al. (2009) believe is due to the presence of dikes with the same NW orientation that the ridge has.

Looking at the features of the gravity map, the marine data show that the more distant gravity anomalies offshore do not have a great influence on the map of La Gomera and its nearest surroundings. So, with the aim of studying the crustal structure of this island, the following interpretation techniques are applied to an area limited up to seven kilometres offshore (Fig. 3). Firstly, the gravity signal of the shallower sources is studied from a local gravity map without the longer wavelength gravity signal, and from the total horizontal gradient of the gravity data. Secondly, the distribution and the geometry of the sources of the gravity field are modelled by a three-dimensional inversion technique.

4. TDX and residual gravity maps

The aim of gravity research is to define the density source distribution in terms of the respective field. However, this is a complex task since the gravity field arises from the integrated effects of a number of sources that the inversion techniques cannot always identify. The resolution of the gravity data and their observation errors, the hypotheses and constraints used in the gravity study, and the inherent limitations of the gravity inversion, mean that sources with superposed effects in the gravity map cannot be modelled correctly. Therefore, a previous analysis of the gravity map using techniques to separate the regional and local components helps to obtain a more complete interpretation of the gravity field. On the one hand, this separation process can be achieved by fitting a polynomial surface to gravity data, which identifies the contribution of the deeper and bigger sources to the gravity map. Then, the residual of this fit corresponds to the effects of local sources. On the other hand, several techniques are available to interpret or to enhance the hidden effects of these sources on the observed gravity map, such as downward continuation, horizontal and vertical derivatives and the high pass filters. These transformed maps help to define the horizontal extension and the borders of the local sources.

In the case of La Gomera, we analyse the gravity signal of the shallower sources removing the regional component of the gravity field fitted by a polynomial surface of degree 2. This regional component is dominated by the gravity maximum located and centred at the north of the island. It could support the hypothesis about the existence of a deep reservoir beneath a volcanic structure, corresponding to the early growth stage of the island. This information will be completed through the following inversion problem. The obtained residual gravity map (Fig. 5a) shows a clear difference between the north and the south of the island, towards where the migration of volcanic activity took place (Ancochea et al., 2003).

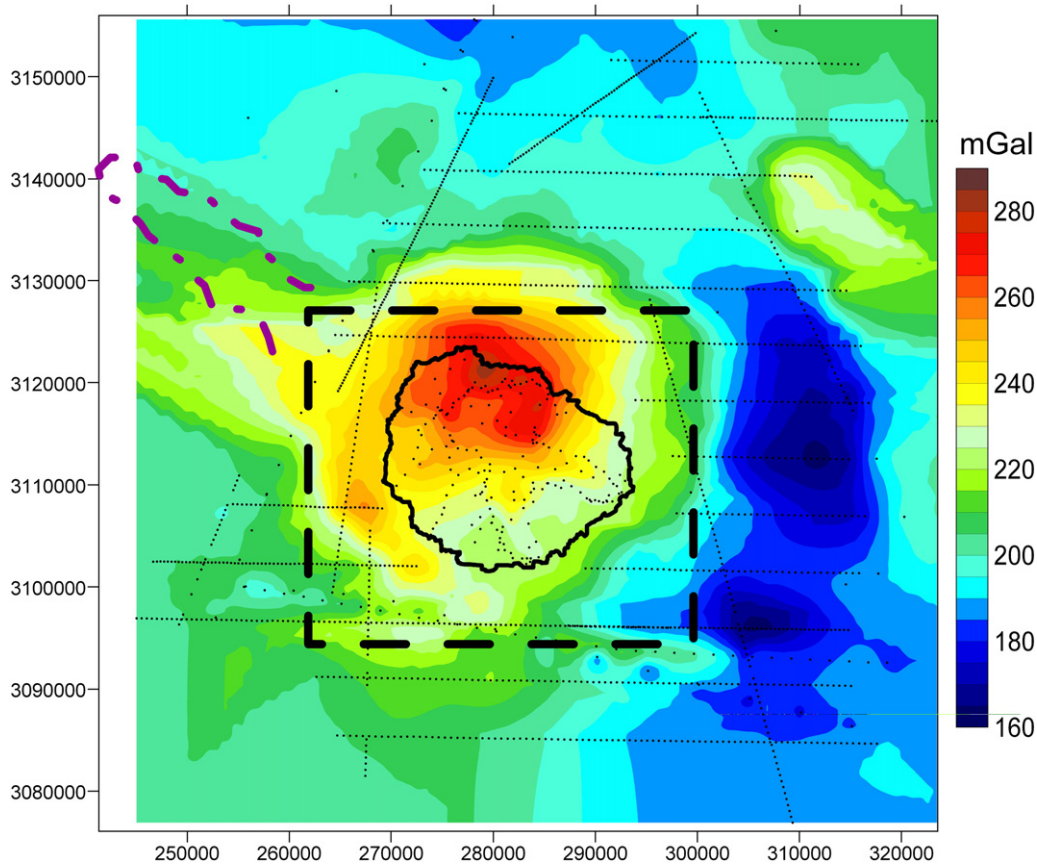

Fig. 3. Map of the Bouguer gravimetric anomalies and distribution of gravity stations. The value used for the density of terrain correction is 2560 kg m^{-3}. The contour interval is 5 mGal. The UTM coordinates are given in metres. The dotted rectangle points out the zone used in the inversion process.

In order to identify the horizontal location and borders of the shallower structures, we have applied the total horizontal derivative TDX of the gravity map (Fig. 5b), which is commonly used as an edge-detection filter (e.g. Cooper and Cowan, 2006). It can be expressed as

$$TDX = \sqrt{\left(\frac{\partial \Delta g}{\partial x}\right)^2 + \left(\frac{\partial \Delta g}{\partial y}\right)^2} \quad (1)$$

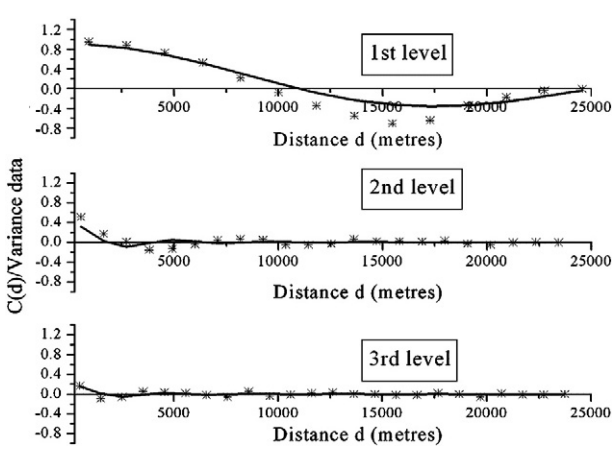

Fig. 4. Covariance functions adjusted (line) to the three successive empirical covariance distributions (stars) used to detect three levels of the gravity signal. The first level is calculated from gravity and marine data and the second and third levels from land data only. From these covariance functions, we estimate the level of accuracy of the gravity map with a standard deviation of 1.77 mGal.

where Δg is the gravity field observed and $\partial\Delta g/\partial x$, $\partial\Delta g/\partial y$ denote the partial derivatives. This map corresponds to a grid of the steepest slopes of the gravity anomaly (i.e. the magnitude of the gradient) at any point on the surface, reflecting the zones where the strong density contrasts occur (positive or negative) for the shallower depths of the crust.

To the north of the island, the highest gravity values of the residual map (Fig. 5a) occupy the zone where the structures of the Basal Complex out crops at surface and ultramaphic plutonic rocks are present (e.g. Rodríguez-Losada and Martínez Frías, 2004). The borders of these structures are identified from numbers 1 to 5 in Fig. 5b. Inside of the Vallehermoso Caldera, the anomalies are associated with the complicated structural features of the caldera (Rodríguez-Losada and Martínez Frías, 2004). In this area, the maximum gravity values are located in the north and decreasing until the south of the caldera (Fig. 5a), skirting round the western rim and the T-PhC complex and cone sheet, which is associated with the minimum gravity values. These maximum gravity areas correspond to two structures (1 and 2 in Fig. 5b). The high residual gravity values inside the caldera towards the western area look like two elongated arms identified by 3 and 6 in Fig. 5b. The complicated geometry exhibited by these structures, located to shallow depths, could be associated to several destructive processes (volcanism and erosion).

In the zones where the residual map exhibits minimum gravity values, the TDX map helps us to establish some considerations. On the one hand, in the northern half of the island, the minimum gravity zones seem to correspond to structures confined by several maximum gravity anomaly areas. These high gravity anomalies are associated with structures 1 and 2 inside the caldera and 3 and 6, in the western area (Fig. 5). Thus, the sources of these minima of gravity can be

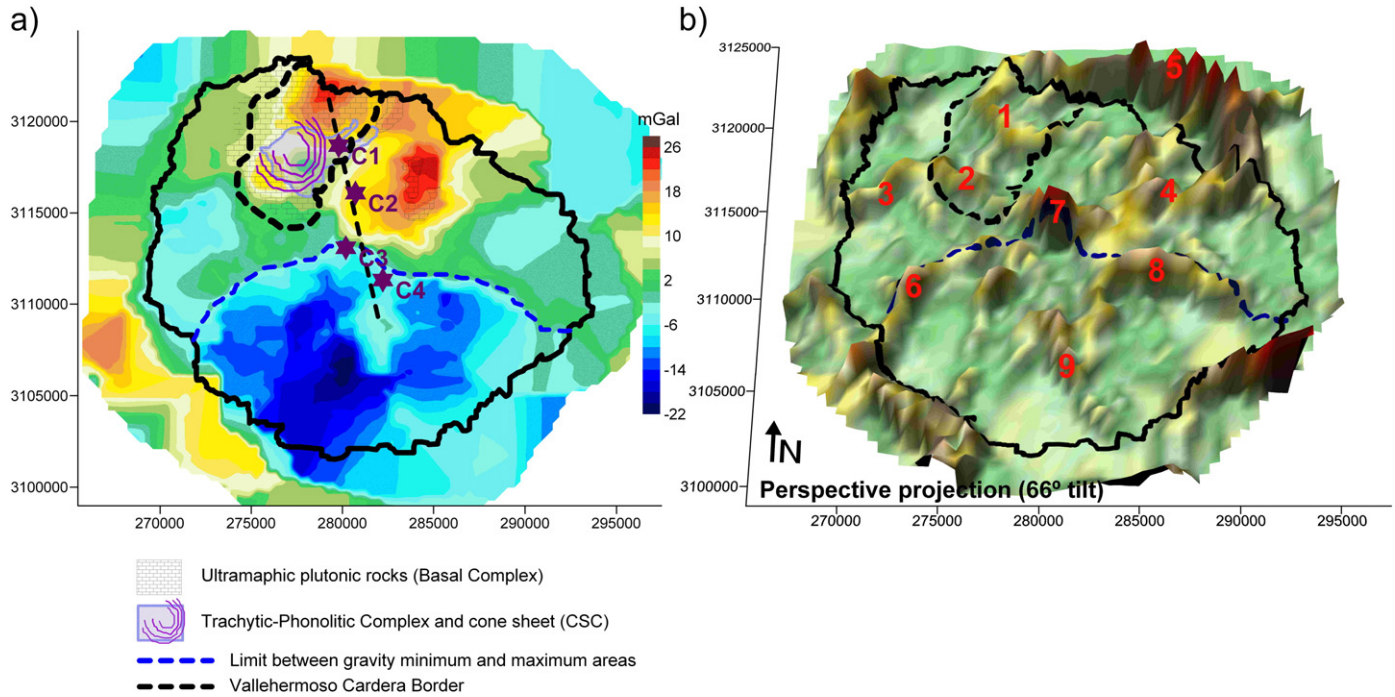

Fig. 5. (a) Residual anomaly map obtained after subtracting a polynomial fit of degree 2 from the gravity map. Several geological features of the Vallehermoso Caldera are included (Rodriguez Losada and Martínez-Frias, 2004): Basal Complex (Ultramaphic plutonic), Trachytic-Phonolitic Complex (T-PhC) and cone sheet, border of the caldera. The stars (C1,...) indicate the distribution of the volcanic centres from Ancochea et al. (2003). (b) Total horizontal derivative of the Bouguer gravity map as a shaded relief map. Blue dotted line divides the island into two zones of maximum and minimum gravity values. The red numbers point out the location of the borders of several shallower structures identified by the maximum values of the TDX. The UTM coordinates are given in metres.

produced by a younger volcanism that alters the oldest structures related to those high gravity values. On the other hand, some structures (6, 7 and 8 in Fig. 5b) are clearly identified on the TDX map where the limit between the minimum and maximum gravity areas is pointed out by the residual map (Fig. 5a). The structural borders denoted by 6, 7, 8 and 9, which are identified in this southern area, could be related to the last stage of the Upper Old Edifice and the feeding roots of the Younger Edifice. This area with the lowest gravity value is just in the zone mainly occupied by the UOL and YN materials.

Paying attention to the reconstruction of the evolution of the main volcanic edifice of La Gomera made by Ancochea et al. (2008), these authors locate four magmatic foci (C1 to C4 in Figs. 1 and 5), which exhibit a migration southwards. The two first inferred volcanic centres (C1 and C2) are the oldest and the composition and thickness of their dikes are similar. They are correlated with the maximum gravity area. The other centres (C3 and C4) are identified by dikes with much younger ages, and they are analogous in thickness and composition. These two younger centres are located at the minimum gravity area. So, the difference between the northern and southern halves of the residual gravity map seems to confirm the possibility of the migration of the volcanism from north to south, relating the maximum and minimum gravity areas with the fact that C1 and C2 dikes are associated with the Old Edifice, whereas C3 and C4 dikes are related to the Young Edifice. The location of C3 coincides just in the area where the TDX maps identify the strongest variation of gravity field (7 in Fig. 5b). This high value of TDX is associated with a strong change in the lateral distribution of the densities of the subsurface structures (it changes from a negative to a positive density contrast) and a shallower location of these density contrasts.

The following inversion process allows a three-dimensional localisation of these structures and a better analysis of sections at different depths.

5. Gravity inversion method

We use the gravity inversion method developed in Montesinos et al. (2005) based on a genetic algorithm (GA). This inversion approach seeks the geometry of the sources of the gravity field, adopting the hypothesis that these structures are characterised by θ mass density contrasts established a priori, according to geologic or geophysical information. Our GA works with a population of individuals (models representing the gravity field sources), which are initially selected at random. Through an iterative evolution process, the population is modified looking for an individual (model) that generates a gravity field fitted to the observed gravity field.

The models are defined from a fixed discrete 3D partition of m prismatic elements of the subsurface volume close to the survey. These prisms are located at a certain depth from the observation surface (topographic surface), taking into account the topographical height of the nearest stations. The model parameters are the densities of each prism of each individual that belongs to the population. Thus, the anomalous sources are constructed assigning one of the θ prescribed density contrasts to each prism through the following evolutive process.

Let us consider that n gravity stations P_i (x_i,y_i,z_i), $i=1,...,n$, are not necessarily gridded, and are located on the topographic surface and with observed gravity anomaly values g_{obs}. The error function determines the adjustment of the calculated gravity from each model to the observed gravity data. This function, which must be minimised is (Montesinos et al., 2005)

$$F(\mathrm{m}_k) = (\mathrm{Am}_k - \mathrm{g}_{obs} - G_k\mathrm{u})^T E_{ss}^{-l} (\mathrm{Am}_k - \mathrm{g}_{obs} - G_k\mathrm{u}) + \beta\, \mathrm{m}_k^T \mathrm{C}_M\, \mathrm{m}_k \quad (2)$$

where A defines the direct problem, E_{ss} is the error matrix of the data, G_k is a constant term calculated as a weighted media residual, u is the unitary vector, β is an appropriate regularisation parameter chosen a

priori, and the m × m matrix C_M is defined describing the uncertainty of the model m_k.

The process starts with a population of np randomly generated models, P(0). The size of the population, np, is selected in function of the quantity of prisms, data, etc., (generally a value of 10 gives successful results, although another selection would not alter the results but the speed of convergence) and it is invariant throughout the process. In the iterative procedure, the individuals of the population are evaluated by the error function in Eq. (2) and can be chosen (selection operator) to continue the evolutive process. Over these selected models, two genetic operators (mutation and cross) act producing ⁄ new individuals, which are also evaluated. Then, the selection operator chooses the best np individuals from these np (old) + ⁄ (new) models. This selection operator is defined by the error function, ensuring that the best models get more copies.

To create new models, the mutation and crossover operators act on these selected individuals. The mutation operator acts on some prisms according to a probability p_m fixed a priori, changing its density contrast value randomly among the θ fixed density contrasts. The cross operator exchanges genes between two individuals with a probability p_c. We use values $p_m = 0.3$ and $p_c = 0.6$, recommended in the GA literature (e.g., Grefenstette, 1986; Michalewicz, 1994; Montesinos et al., 2005). The iterative procedure is repeated until a model that minimises the error function is found or the data misfit reaches a predetermined value according to the estimated error for the gravity map (end process criterion).

The non-uniqueness of the solution of the gravity inverse problem and the ambiguity due to the errors of gravity data, together with their discrete and irregular distribution and the suitable (or unsuitable) choice of the density contrasts (positive and negative) must be reduced including some constraints in the process. Therefore, to dilute this uncertainty in the solution of the model, we employ a smoothing technique equivalent to minimising the whole anomalous mass (Silva et al., 2001). This operator acts on each model when the error function reaches the minimum value, assigning an average value of the density contrasts of the adjacent prisms to each prism. Then, these new smoothed individuals are evaluated again. If the best model fits the observed gravity data successfully according to the fixed limit, or it does not improve the previous models after several stages, the inversion ends. Otherwise, the evolutive process happens again.

To apply this methodology to study the crustal structure of La Gomera, we adopted a 3D partition of the local subsoil into 14,768 rectangular prisms. They depart from 1000 m below the observation surface (this depth corresponds to 460 m b.s.l., taking into account a mean topographic height of 540 m), and have sides ranging from 900 m for the shallowest elements to 1200 m for the deepest blocks (at 13,500 m depth b.s.l.). The shallowest and deepest depth of the model from the observation surface is selected taking into account that the distance between contiguous stations is approx. 1000 m and the mean diameter of the survey is about 35,000 m.

For the whole subsoil volume, we have selected the possible density contrasts by means of the extreme values, -400 kg m^{-3} and $+400$ kg m^{-3}, and an exponential variation law of these contrasts. These values have been selected empirically to obtain connected and well-developed volumes (contrasts that are too low produce connected, convex and rounded volumes; contrasts that are too high produce disaggregated and "skeleton" volumes). Although there are no any measurements of density rocks in the area, these density contrast values are similar to other ones used in the gravity studies achieved on various islands of this archipelago with an analogous volcanic environment (e.g., Araña et al., 2000; Camacho et al., 2001, 2009; Montesinos et al., 2005, 2006; Gottsmann et al., 2008).

The distribution of density contrasts obtained through the inversion of the gravity field is always obtained without considering any horizontal stratification of layers, which are not identified by the gravity data. So, it is not possible to obtain absolute density data without falling into a simple hypothesis when we use only gravity techniques. A detailed seismic study would help to solve this matter.

The model of anomalous structures found for several depths with respect to the sea level (although the model is made from the topographic surface) is shown in horizontal and vertical sections (Figs. 6 and 7). The final inversion residuals have a mean square root value of 1.24 mGal. This value is near to the estimated error of the data by the covariance analysis, which is assumed as indicative of the inner coherence of the gravity data (land and marine). These residuals correspond to very local gravity disturbances, local inaccuracies of the topographical correction due to the use of a mean density for the terrain, etc.

6. Results and discussion

The crustal model of La Gomera obtained from the inversion of the gravity field is interpreted with regard to the geological and volcanological studies mentioned in previous sections, which allow us to point out the reliability of the results. Moreover, the gravity inversion defines, in a 3D context, the bodies whose borders were identified in the TDX map, completing the information provided from the former analysis.

The main source of the gravity field modelled is a high density body, which appears from the deepest sections of the model (Figs. 6 and 7), showing its relevance in the building of La Gomera and its independence of other structures associated with neighbouring islands. The deepest sectors of this body correspond to the earlier stage of submarine growth of La Gomera (SB), related to the ancient extrusion of massive basaltic material that forms the main and oldest structure of the island, the Basal Complex. This high density body seems to be a unique quasi-circular structure in the deepest sections and it extends towards shallower sections where it continues with a more irregular geometry.

Taking into account that the construction of the following edifice (Old Edifice) took place upon the remainder of this SB, the successive sections of the gravity inversion model provide information about the original morphology and location of this volcanic unit. So, the main high density structure of the model, from the deepest sections up to nearly $z = -3000$ m, occupies the area where Ancochea et al. (2006) identify the composite volcano that corresponds to the LOL, which is defined as slightly bigger than its current geometry and centred towards the north of the island (red dotted line in Fig. 6). The model reveals the coincidence between this high density structure and the northern rim of this suggested emergent edifice, 5000 m off the present northern coastline (Figs. 1 and 6). However, the structure associated with this old volcano appears smaller than the previous geological studies have established. This body has a mass of 0.449×10^{15} kg and its centre of mass is located at 6000 m b.s.l. Its horizontal sections have an average diameter of about 20,000 m, with a quasi-circular structure extending from 9000 m to 4000 m b.s.l. From this depth to the top of the model, the spherical geometry of the body is broken. This fact could be probably due to the important role of the destructive processes in the evolution of the island, including failures of the NW and NE flanks, which must have been destroyed in several intense events. It should be noted that the northern portion of this body is modelled with the influence of marine gravity data and, therefore, there is a lower confidence with respect to the southern portion controlled from onshore gravity survey. Another interesting feature of this body is observed in the b-profile shown in Fig. 7. From about 4000 m deep, until it reaches the surface, its southern edge is particularly straight with a slope of 60° suggesting some ancient tectonic feature.

In the shallowest sections of the model, the prolongation of the main gravity field source is modelled as several units reaching the surface, just in the zones where the BC outcrops (Fig. 1 and A, B and C structures in Fig. 7), mainly inside the area of the Vallehermoso

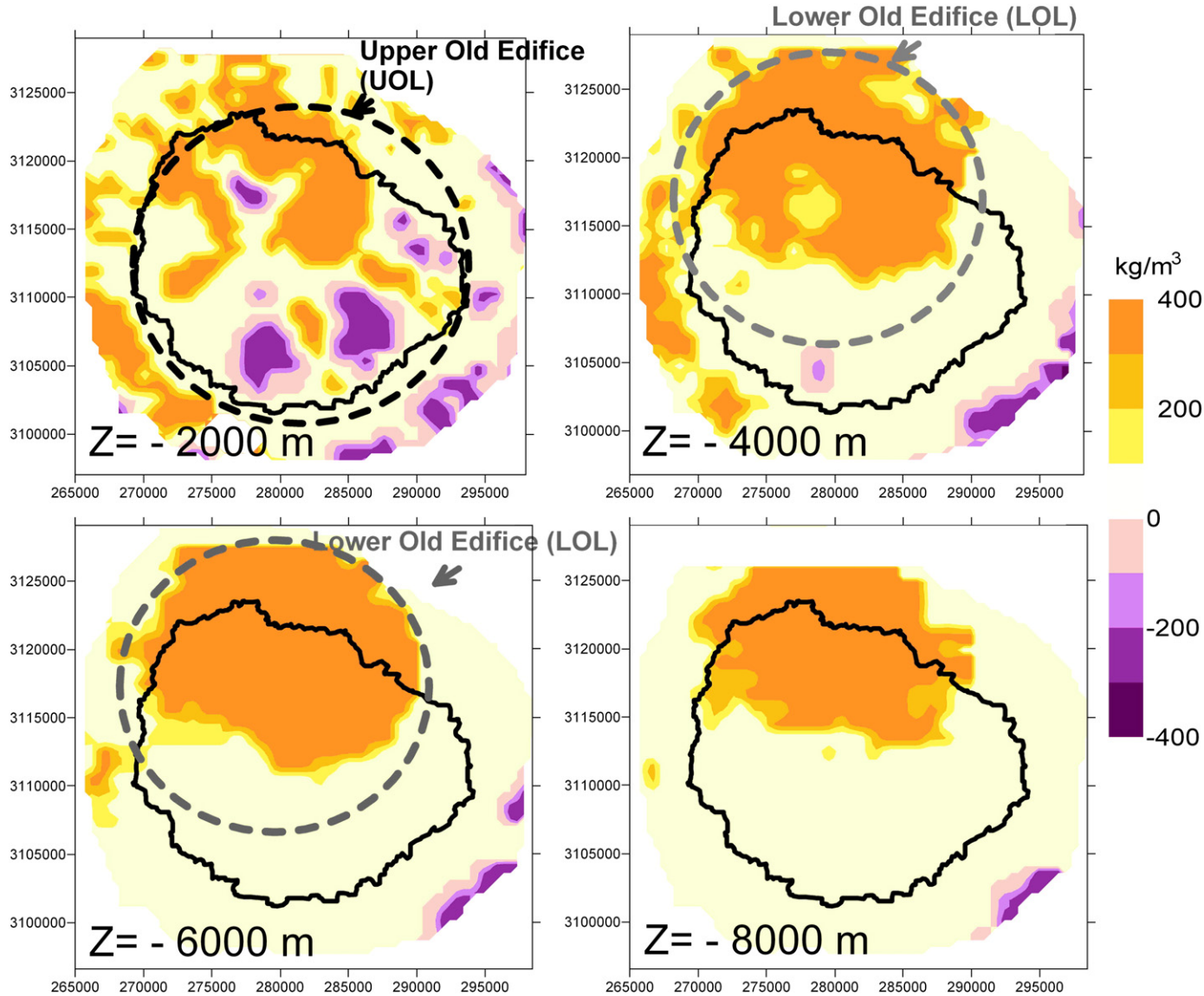

Fig. 6. Adjusted model of density contrasts obtained by means of 3D gravity inversion. The density contrast values used in the model correspond to a discrete set of values between −400 and 400 kg/m^3. Several horizontal sections, from 2000 to 8000 m depth, are shown. At greater depths the model does not identify any structure. The location of the basaltic volcano corresponding to LOL (red circle) and the UOL (black circle) are indicated in two sections. The UTM coordinates are given in metres.

Caldera. From the TDX maps these structures were observed, although its connection to deeper sources was not reflected.

The unit marked by C is a large body in the area of Hermigua (Figs. 6 and 7), where Cendrero (1971) locates the "Lower Old Basalts" and the polygenic volcanic breccias, which represent the earlier stage in the construction the Lower Old Edifice (LOL). Another high density structure, such as D (in the Alojera area) (Figs. 1 and 7), could be associated with the plumbing system in the growth stages of this Old Edifice.

The UOL corresponds to emissions that spread out much further south than the area occupied by the preceding LOL. In this case, the model does not identify a unique structure related to the existence of a wide composite volcano, which would have risen over the former edifice (Ancochea et al., 2006). As the sections of the model are shallower, several gravity sources appear to also be distributed in the southern half, the E (near to Valle de Gran Rey), F (La Laja) and G (Benchijigua) bodies (Fig. 1, 6 and 7) supporting the hypothesis about a migration of the activity towards the southern area. These smaller high density structures could represent the feeding roots of the first activity of the UOL at these sectors, where this volcanic unit consists mainly of basaltic rocks (Ancochea et al., 2006). The southernmost high density structure, G body (Fig. 7b), could be interpreted as an elongation towards the south of the main high density body of the model as well as the bodies E and F (Fig. 7c). This fact suggests the possible change of magmatic activity towards the south of the island.

Considering the low gravity areas, several sources are modelled. They correspond to structures with a similar pattern, that is, shallow and elongated bodies. Thus, in the eastern sector of the island, the model identifies two low-density structures (I, J in Fig. 7) located at the Jaragán area (see Fig. 1). These structures have a mass of about 0.55×10^{13} kg and its centre of mass is located about 1200 m deep (1200 m b.s.l. corresponds to a mean of 2000 m below the surface of observation), showing an elongated geometry. Another small body, M in Fig. 7, appears at the shallowest sections of the model and with a similar linear structure as the former. These shallow locations and their geometry do suppose that these structures can be interpreted to represent the accumulation of low density material associated with the shallow extrusive during the UOL and YN stages. In this last stage, the lava flows erupted from the central area of the island, spread all over, covering a vast extension and filling deep valleys in the central

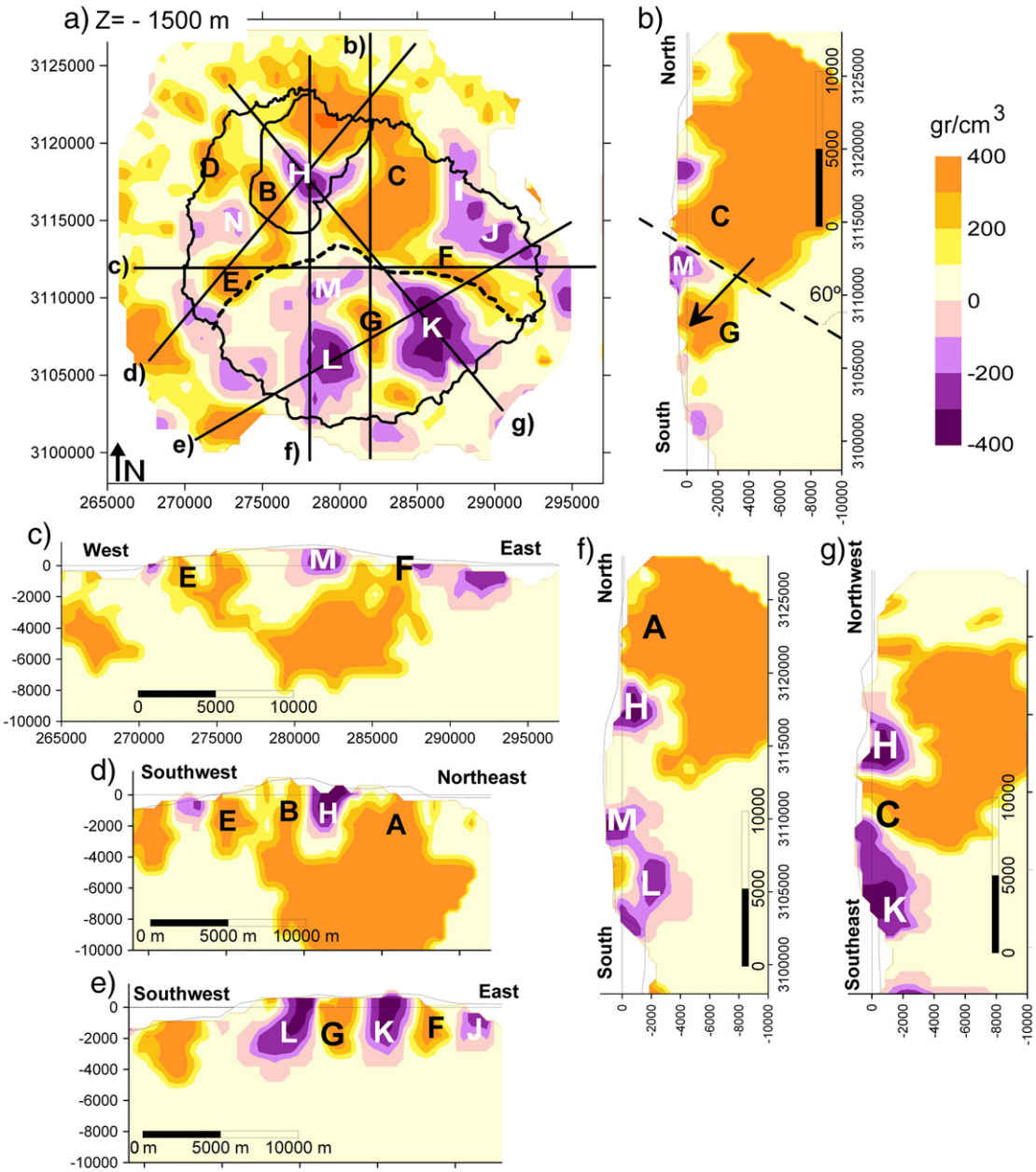

Fig. 7. a) Shallowest section of the adjusted model of density contrasts obtained by means of 3D gravity inversion to 1500 m depth. Several structures mentioned in the text are indicated by capital letters (black letters: high density bodies; and white letters: low density bodies). The division line between the low and high gravity zones is also included. The black lines indicate the vertical profiles of the model shown in (b), (c), (d), (e), (f) and (g). In profile (b), the possible tectonic lineament (dotted line) and the progression of volcanic activity (black arrow) are indicated. The UTM coordinates are given in metres.

area, reaching out even to the periphery of the island. Moreover, intense erosional processes and a probable new catastrophic landslide took place (Ancochea et al., 2006). These processes could be the origin of these areas that exhibit low density material with respect to the high density structures corresponding to the earlier OL.

In the same way, at the northwestern of the island, another shallow and low density body, N in Fig. 7a, appears surrounded by high density structures, not far from the zone where a collapse is conjectured. This body is surrounded by high density structures and could be justified by the destructive processes which affect this area.

With regard to the Vallehermoso Caldera, it is possible to obtain some interesting information from the gravity inversion model. In this area, where an essential part of the felsic activity of the UOL is comprised of several volcanic systems and dike swarms, the inversion model identifies low and high density structures. The interpretation of this model can be made by also taking into account the previous geological studies which suggest several possibilities for the origin of this caldera.

The modelling of the subsurface structure of Vallehermoso Caldera from the gravity inversion is shown through several profiles (Fig. 7d, f and g). The combination of the results obtained from the gravity field investigations and previous geological studies seem to provide indications for a vertical collapse origin of this caldera, which is also affected by other destructive processes (erosion and landslides) as are suggested by other authors (Ancochea et al., 2006; Paris et al., 2005). There are several features of the inversion model which justify this point of view.

The region inside this caldera where the centre of a cone sheet swarm and the Eastern Radial Swarm are located (CSC and ERS in Fig. 1), is occupied by a low-density structure (H in Fig. 7). This body

has a mass of 0.594×10^{13} kg, its centre of mass is located nearly −1100 m b.s.l., and is surrounded by high density structures (A and B bodies inside the caldera border, C body outside of the caldera), except in the southeastern and northwestern zones corresponding to shallowest sections of the model (Fig. 7a, f and g respectively). These high density structures, as previously mentioned, are located in the areas where the part of the Basal Complex crops out rimming the south, west and northern limits of the caldera. These bodies delineate a ring around the low density body H. This fact seems to support the hypothesis about the existence of an older edifice affected by vertical collapse, the remnants of this old edifice being the high density structures (Montesinos et al., 2003; Gottsmann et al., 2008).

Peripheral normal faults and fractures, identified by Rodríguez-Losada and Martínez Frías (2004) (see Fig. 2 of these authors), are located just between the high and low density bodies B and H. These fractures are joined to the vertical morphology observed in Fig. 7d in agreement with the subsurface architecture, which could be associated with the idea of a vertical collapse occurring in this area. Moreover, from the vertical profile it is possible to find a neutral density material which borders the body H and also extends to a considerable depth (to 4000 m b.s.l., see Fig. 6), significantly displacing the underlying high density body. This displacement would require a deep structural perturbation unlikely to be induced by a superficial landslide. The perturbation is consistent with a down faulting of a sizable part of the high density interior of the LOL, and a possible alteration in the form of lateral landslides directed to the north.

There are several possibilities to explain the low-density body H. On one hand, this zone is dominated by NW trachy-phonolitic formations (Ancochea et al., 2003; Rodríguez-Losada and Martínez Frías, 2004), which consist of intrusive materials (phonolitic domes, trachytic dikes, and small syenitic bodies), breccias, tuffs and massive peripheral lava flows. The intrusive swarm is composed of a cone sheet and two radial complexes (Ancochea et al., 2003; Rodríguez-Losada and Martínez Frías, 2004). Thus, these materials which fill this embayment and the avalanche debris deposits along the Vallehermoso ravine correspond to this low-density body modelled from the gravity inversion. On the other hand, the existence of highly hydrothermally altered areas in the felsic rocks is a feature that defines the T-PhC and the Vallehermoso Caldera (e.g. Rodríguez-Losada and Martínez Frías, 2004). In the same way as is suggested for the Las Cañadas Caldera on the island of Tenerife (Gottsmann et al., 2008), the hydrothermal alteration of the volcanic complex in the Vallehermoso area and the erosion can cause secondary void space leading to lower bulk densities. Assuming that at least parts of the interior of the complex have undergone a similar degree of alteration, one would expect lower densities for the interior of the complex. Perhaps this feature is best exemplified by the conduit-shaped low-density structure beneath T-PhC, which is extended upwards and to the northeast and northwest from the centre of the low density (Fig. 7a and g).

Regarding the more recent growth stage of La Gomera, the YN emitted lava flows that covered the centre and the south of the island completely whilst, on the northern flanks, these flows only filled ravines that had already been excavated. Therefore, it can be supposed that the main emission centres would be situated in the southern half of the island, where the effects were more evident. For these two reasons, the low-density structures M, L and K (Fig. 7) located in the southern half of the model could be linked to the feeding system of the Younger Edifice. The low density bodies K and L do not exhibit any connection between each other, supporting the idea about the existence of several stages related to the last period of activity on the island. The body K is modelled by a low-density structure of 1.53×10^{13} kg and the centre of mass is located at a depth of −1191 m b.s.l. The body L is located deeper, with its centre of mass at −1776 m b.s.l., being a structure smaller than the body K, with a mass of 1.05×10^{13} kg. The body M is the shallowest, with an elongated form and connected with the most important L body. This location and configuration could be associated with the process of filling deep ravines, which took place during the YN stage.

Finally, it is possible to make some comments related to the framework of the Canary Archipelago. With this gravity inversion model, the gravity studies and the corresponding models of all islands of this archipelago are already available. From these results, we can reveal some general comparative features. As has been established by many authors (e.g., Marinoni and Pasquare, 1994; Anguita and Hernán, 2000; Dañobeitia and Canales, 2000) the islands of this archipelago rose up as independent blocks or groups of blocks. The gravity inversion model shows the deep structure of La Gomera is independent of the neighbouring islands, evolving its growth stages without direct connection to the volcanic activity suffered on other islands. However, the same pattern is observed as in the crustal model obtained by gravity inversion on all islands of Canaries (e.g. Araña et al., 2000; Montesinos et al., 2005). In all of them, a main deep high density body is identified in the model. These bodies constitute the hypabyssal roots of the early structure of each island in its submarine growth stage. The following growth stages are associated to several structures, which are identified in the respective models (see corresponding references). Moreover, it is possible to establish a clear analogy among the models obtained for La Gomera, Fuerteventura (Montesinos et al., 2005) and La Palma (Camacho et al., 2009), which are the islands of the archipelago where the Basal Complex outcrops on the surface.

7. Conclusions

We hereby present a new Bouguer gravity anomaly map of La Gomera island using land and marine gravity data. The study of this map provides new information compared to the previously published works from geologic and volcanologic data (Ancochea et al., 2003, 2006, 2008; Rodríguez-Losada and Martínez Frías, 2004; Paris et al., 2005; Llanes et al., 2009; etc.), offering a remarkable view of the crustal structures from shallower sections down to depths about 9000 km.

The quality of the gravity value and the homogeneization of data sets are studied, estimating a standard deviation of 1.77 mGal of non-correlated noise, which is quite enough for this study.

Firstly, this gravity map is analysed through its residual component, which corresponds to sources whose borders are identified by the total horizontal derivative technique. This analysis is a useful tool to show features of shallower structures that could be related to younger volcanic episodes, which altered the first kilometres of the crust. From the residual and derivative maps are exposed structures related to the last stage of the Upper Old Edifice and the feeding roots of the Younger Edifice. Among these structures, the maps reveal complicated features inside of the Vallehermoso Caldera, which are associated to several destructive processes which took place in this area. Moreover, it is deduced a clear correlation between the residual gravity and the results obtained by other geological and volcanological studies about the volcanism in the island and its progression toward the southern area of the island. This study supports this southern migration of the volcanism.

Secondly, applying a gravity inversion technique based on a genetic algorithm (Montesinos et al., 2005), we have constructed a crustal model by limiting the possible density contrasts between −400 and 400 kg m^{-3}. The gravity data calculated for this model adjust the observed data with differences following a Normal distribution, with a standard deviation of 1.24 mGal according to the level of noise estimated for the gravity data.

The use of gravity modelling in La Gomera constitutes an advance in the study of this volcanic island where the deep crust is unknown. The inversion of the gravity field provides a model of its sources,

which supply information about some volcanic processes, the oldest stages of the grown of the island and about structures which have been suggested by surface geological inspection. Thus, from these results are possible to support or to question some theories and hypothesis argued by other geological or geophysical studies.

The main advantage of the inversion method used is its three-dimensional context, modelling structures with complex geometry which are common in volcanic zones. Moreover, it is only necessary the use of limits of density contrasts as a priori information. A different selection of these values would not alter fully the solution: the general information would be similar and aspects related with the total volume would change.

The results of the inversion process show a main intrusive body related to the earlier stage of submarine growth of La Gomera. This body is associated to the ancient extrusion of massive basaltic material that forms the main and oldest structure of the island, the Basal Complex, which reaches the surface. The features of this deep structure are deduced from this inversion process.

The location of the main high density structures in the shallower sections of the model coincides with the area occupied by the composite volcano (LOL) identified by Ancochea et al. (2006). The original morphology or the precise location of the OL, which it was not possible to determine from geological information, is modelled in this study. Its irregular geometry shows the important role of the destructive processes in this phase of evolution. Moreover, other high density structures found in the model are associated with this edifice. The existence of a second great volcano in this period, UOL, which is identified by this author, is not supported by the gravity inversion.

The following period of volcanic activity on the island was complicated with different emission centres, which coexisted. This activity can be related to high and low-density structures according to the different activity. Furthermore, as the previous analysis established, the several structures identified inside the Vallehermoso caldera could be associated with a vertical collapse and other destructive processes, which resulted in the present caldera and that were already studied by other authors (Rodriguez-Losada and Martinez Frías, 2004).

These results show the usefulness of the gravity inversion method to provide information about the growth and evolution of La Gomera. The model of the crustal structures obtained completes the geological and volcanological information, giving valuable details about the possible origin and feeding systems of the volcanic units observed on the surface.

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